

Exercise 2.9.2 Obtain the geodesic equations (2.87) in three different ways:

- (a) By using the Euler-Lagrange equations (and $[g^{\mu\nu}]$ from Exercise 1.
- (b) By extracting $[g^{\mu\nu}]$ from the line element (2.86), calculating the $\Gamma_{\nu\sigma}^{\mu}$ from $g^{\mu\nu}$ and $g_{\mu\nu}$ using equation (2.13), and then calculating $\ddot{x}^{\mu}$ using equation (2.71).
- (c) By substituting for T, X, Y, Z in $\ddot{T} = \ddot{X} = \ddot{Y} = \ddot{Z} = 0$, using equations (2.85)

Solution. Let K be an inertial (non-rotating) system with coordinates (T, X, Y, Z) and line element

$$c^2 d\tau^2 = c^2 dT^2 - dX^2 - dY^2 - dZ^2 \quad (2.84)$$

Denote $X^0 = cT$, $X^1 = X$, $X^2 = Y$, $X^3 = Z$.

Let K' be a rotating system with coordinates (t, x, y, z) , defined by

$$\begin{aligned} t &= T & T &= t \\ x &= X \cos \omega t + Y \sin \omega t & X &= x \cos \omega t - y \sin \omega t \\ y &= X \sin \omega t + Y \cos \omega t & Y &= x \sin \omega t + y \cos \omega t \\ z &= Z & Z &= z \end{aligned} \quad (2.85)$$

Denote $x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$, and $\dot{x}^0 = c \dot{t}$, $\dot{x}^1 = \dot{x}$, $\dot{x}^2 = \dot{y}$, $\dot{x}^3 = \dot{z}$, $\ddot{x}^1 = \ddot{x}$, $\ddot{x}^2 = \ddot{y}$, $\ddot{x}^3 = \ddot{z}$, where dot represents differentiation by τ . The spacetime geodesic equations of motion are

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma_{\nu\sigma}^{\mu} \frac{dx^{\sigma}}{d\tau} \frac{dx^{\nu}}{d\tau} = 0 \quad (2.71)$$

We are to show that for a free particle with mass in the rotating frame K' , these equations may be expressed as

$$\begin{aligned} \ddot{t} &= 0 \\ \ddot{x} - \omega^2 x \dot{t}^2 - 2\omega \dot{y} \dot{t} &= 0 \\ \ddot{y} - \omega^2 y \dot{t}^2 + 2\omega \dot{x} \dot{t} &= 0 \\ \ddot{z} &= 0 \end{aligned} \quad (2.87)$$

Part (a): The Euler-Lagrange equations (2.17) are $\frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^{\sigma}} - \frac{\partial L}{\partial x^{\sigma}} = 0$ where the

Lagrangian is $L \stackrel{(2.1.d)}{=} \frac{1}{2} g_{\mu\nu} \dot{x}^{\mu} \dot{x}^{\nu}$ and we regard x^{μ} and $\dot{x}^{\mu}$ as independent variables. Keep in mind that $g_{\mu\nu}$ is a function of x^{μ} .

First,

$$\dot{x}^0 = c \dot{t} = \frac{c dt}{d\tau} \stackrel{(A.6)}{=} \frac{c}{\gamma}, \text{ a constant} \Rightarrow \ddot{x}^0 = c \ddot{t} = 0 \Rightarrow \ddot{t} = 0 \quad \checkmark$$

In terms of K' coordinates, the line element is given as

$$\begin{aligned} g_{\mu\nu} dx^\mu dx^\nu &= c^2 d\tau^2 \\ &= [c^2 - \omega^2(x^2 + y^2)] dt^2 + 2\omega y dx dt - 2\omega x dy dt - dx^2 - dy^2 - dz^2 \end{aligned} \quad (2.86)$$

From the line element we easily see that

$$\begin{aligned} g_{00} &= \frac{c^2 - (x^2 + y^2) \omega^2}{c^2}, \quad g_{11} = -1, \quad g_{22} = -1, \quad g_{33} = -1, \\ g_{01} &= g_{10} = \omega y, \quad g_{02} = g_{20} = -\omega x, \quad \text{and all others are zero.} \end{aligned}$$

Thus, the Lagrangian, L , is

$$\begin{aligned} L &= \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &= \frac{1}{2} \left\{ \frac{c^2 - (x^2 + y^2) \omega^2}{c^2} c^2 \dot{t}^2 + 2 y \omega \dot{x} \dot{t} - 2 x \omega \dot{y} \dot{t} - \dot{x}^2 - \dot{y}^2 - \dot{z}^2 \right\} \\ &= \frac{1}{2} [c^2 - (x^2 + y^2) \omega^2] \dot{t}^2 + \omega (y \dot{x} - x \dot{y}) \dot{t} - \frac{1}{2} (\dot{x}^2 - \dot{y}^2 - \dot{z}^2) \end{aligned}$$

So,

$$\frac{\partial L}{\partial x^1} = \frac{\partial L}{\partial x} = -\omega^2 x \dot{t}^2 - \omega \dot{y} \dot{t}$$

$$\frac{\partial L}{\partial x^2} = \frac{\partial L}{\partial y} = -\omega^2 y \dot{t}^2 + \omega \dot{x} \dot{t}$$

$$\frac{\partial L}{\partial x^3} = \frac{\partial L}{\partial z} = 0$$

$$\frac{\partial L}{\partial \dot{x}^1} = \frac{\partial L}{\partial \dot{x}} = \omega y \dot{t} - \dot{x}$$

$$\frac{\partial L}{\partial \dot{x}^2} = \frac{\partial L}{\partial \dot{y}} = -\omega x \dot{t} - \dot{y}$$

$$\frac{\partial L}{\partial \dot{x}^3} = \frac{\partial L}{\partial \dot{z}} = -\dot{z}$$

$$\begin{aligned}\frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^1} &= \omega \dot{y} \dot{t} - \ddot{x} \\ \frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^2} &= -\omega \dot{x} \dot{t} - \ddot{y} \\ \frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^3} &= -\ddot{z}\end{aligned}$$

and Euler-Lagrange equations $\frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{x}^\sigma} \right) - \frac{\partial L}{\partial x^\sigma} = 0$ become:

$$\begin{aligned}(1) \quad 0 &= \omega \dot{y} \dot{t} - \ddot{x} + \omega^2 x \dot{t}^2 + \omega \dot{y} \dot{t} \Rightarrow \ddot{x} - \omega^2 x \dot{t}^2 - 2 \omega \dot{y} \dot{t} = 0 \quad \checkmark \\ (2) \quad 0 &= -\omega \dot{x} \dot{t} - \ddot{y} + \omega^2 y \dot{t}^2 - \omega \dot{x} \dot{t} \Rightarrow \ddot{y} - \omega^2 y \dot{t}^2 + 2 \omega \dot{x} \dot{t} = 0 \quad \checkmark \\ (3) \quad 0 &= -\ddot{z} \quad \checkmark\end{aligned}$$

Part (b): $g_{\mu\nu}$ were found in Part (a). In matrix form this becomes

$$(g_{\mu\nu}) = \begin{pmatrix} 1 - \frac{(x^2+y^2)\omega^2}{c^2} & \frac{y\omega}{c} & -\frac{x\omega}{c} & 0 \\ \frac{y\omega}{c} & -1 & 0 & 0 \\ -\frac{x\omega}{c} & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Using Mathematica, we then find the contravariant metric tensors:

$$(g^{\mu\nu}) = (g_{\mu\nu})^{-1} = \begin{pmatrix} 1 & \frac{y\omega}{c} & -\frac{x\omega}{c} & 0 \\ \frac{x\omega}{c} & -1 + \frac{y^2\omega^2}{c^2} & -\frac{xy\omega^2}{c^2} & 0 \\ -\frac{x\omega}{c} & -\frac{xy\omega^2}{c^2} & -1 + \frac{x^2\omega^2}{c^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

The Christoffel coefficients are computed from the covariant and contravariant metric tensors. There are 64 connection coefficients:

$$\Gamma_{\nu\sigma}^\mu = \frac{1}{2} g^{\mu\rho} (\partial_\nu g_{\rho\sigma} + \partial_\sigma g_{\nu\rho} - \partial_\rho g_{\nu\sigma}) = g^{\mu\delta} \Gamma_{\delta\nu\sigma}.$$

Using Mathematica, we find that the only non-zero ones are:

$$\begin{aligned}\Gamma_{00}^1 &= -\frac{x\omega^2}{c^2}, \quad \Gamma_{02}^1 = \Gamma_{20}^1 = -\frac{\omega}{c}, \\ \Gamma_{00}^2 &= -\frac{y\omega^2}{c^2}, \quad \Gamma_{01}^2 = \Gamma_{01}^2 = \frac{\omega}{c}.\end{aligned}$$

For reference, the non-zero coefficients are computed from $g^{\mu\delta}$ and the non-zero $\Gamma_{\delta\nu\gamma}$ coefficients, as shown below. Take care in the computations for $\Gamma_{\delta\nu\gamma}$ that terms like $\frac{\partial x}{\partial t}$ equal zero because the coordinates are considered to be independent of each other.

μ	δ	$g^{\mu\delta}$
0	0	1
0	1	$\frac{y\omega}{c}$
0	2	$-\frac{x\omega}{c}$
0	3	0
1	0	$\frac{y\omega}{c}$
1	1	$-1 + \frac{y^2\omega^2}{c^2}$
1	2	$-\frac{xy\omega^2}{c^2}$
1	3	0
2	0	$-\frac{x\omega}{c}$
2	1	$-\frac{xy\omega^2}{c^2}$
2	2	$-1 + \frac{x^2\omega^2}{c^2}$
2	3	0
3	0	0
3	1	0
3	2	0
3	3	-1

&

δ	ν	γ	$\Gamma_{\delta\nu\gamma}$
0	0	1	$-\frac{x\omega^2}{c^2}$
0	0	2	$-\frac{y\omega^2}{c^2}$
0	1	0	$-\frac{x\omega^2}{c^2}$
0	2	0	$-\frac{y\omega^2}{c^2}$
1	0	0	$\frac{x\omega^2}{c^2}$
1	0	2	$\frac{\omega}{c}$
1	2	0	$\frac{\omega}{c}$
2	0	0	$\frac{y\omega^2}{c^2}$
2	0	1	$-\frac{\omega}{c}$
2	1	0	$-\frac{\omega}{c}$

 $\Rightarrow$

μ	ν	γ	$\Gamma_{\nu\gamma}^{\mu}$
1	0	0	$-\frac{x\omega^2}{c^2}$
1	0	2	$-\frac{\omega}{c}$
1	2	0	$-\frac{\omega}{c}$
2	0	0	$-\frac{y\omega^2}{c^2}$
2	0	1	$\frac{\omega}{c}$
2	1	0	$\frac{\omega}{c}$

This yields the geodesic equations (2.87): $\ddot{\mathbf{x}} + \Gamma_{\nu\sigma}^{\mu} \dot{x}^{\nu} \dot{x}^{\sigma} = 0$.

$$\begin{aligned}
 0 &= \ddot{x}^0 + \Gamma_{\nu\sigma}^0 \dot{x}^{\nu} \dot{x}^{\sigma} = c \ddot{t}^0 + 0 = c \ddot{t} \Rightarrow 0 = \ddot{t} \quad \checkmark \\
 0 &= \ddot{x}^1 + \Gamma_{\nu\sigma}^1 \dot{x}^{\nu} \dot{x}^{\sigma} = \ddot{x}^1 + \Gamma_{00}^1 (\dot{x}^0)^2 + 2 \Gamma_{02}^1 \dot{x}^0 \dot{x}^2 \\
 &= \ddot{x}^1 + \Gamma_{00}^1 c^2 \dot{t}^2 + 2 \Gamma_{02}^1 c \dot{y} \dot{t} \\
 &= \ddot{x} - \left(\frac{x \omega^2}{c^2} \right) c^2 \dot{t}^2 + 2 \left(-\frac{\omega}{c} \right) c \dot{y} \dot{t} = \ddot{x} - \omega^2 x \dot{t}^2 - 2 \omega \dot{y} \dot{t} \quad \checkmark \\
 0 &= \ddot{x}^2 + \Gamma_{\nu\sigma}^2 \dot{x}^{\nu} \dot{x}^{\sigma} = \ddot{x}^2 + \Gamma_{00}^2 (\dot{x}^0)^2 + 2 \Gamma_{01}^2 \dot{x}^0 \dot{x}^1 \\
 &= \ddot{x}^2 + \Gamma_{00}^2 c^2 \dot{t}^2 + 2 \Gamma_{01}^2 c \dot{x} \dot{t} \\
 &= \ddot{y} - \frac{\omega^2}{c^2} y c^2 \dot{t}^2 + 2 \frac{\omega}{c} c \dot{x} \dot{t} = \ddot{y} - \omega^2 y \dot{t}^2 + 2 \omega \dot{x} \dot{t} \quad \checkmark \\
 0 &= \ddot{x}^3 + \Gamma_{\nu\sigma}^3 \dot{x}^{\nu} \dot{x}^{\sigma} = \ddot{x}^3 + 0 = \ddot{z} \quad \checkmark
 \end{aligned}$$

Part (c): We can rewrite equation (2.84) by dividing both sides by $d\tau^2$:

$$c^2 \stackrel{(2.84)}{=} \dot{c}^2 \dot{T}^2 - \dot{X}^2 - \dot{Y}^2 - \dot{Z}^2.$$

So,

$$\begin{aligned}
 0 &= \dot{c}^2 = 2[c^2 \dot{T} \ddot{T} - \dot{X} \ddot{X} - \dot{Y} \ddot{Y} - \dot{Z} \ddot{Z}] \\
 \Rightarrow \quad \ddot{T} &= \ddot{X} = \ddot{Y} = \ddot{Z} = 0.
 \end{aligned}$$

Thus, $\ddot{t} \stackrel{(2.85)}{=} \ddot{T} = 0$ and $\ddot{z} \stackrel{(2.85)}{=} \ddot{Z} = 0 \quad \checkmark$

From the 2nd equation on RHS of equations (2.85) we get

$$\dot{X} = -\omega x \dot{t} \sin \omega t - \omega y \dot{t} \cos \omega t + \dot{x} \cos \omega t - \dot{y} \sin \omega t.$$

Then, keeping in mind that $\ddot{t} = 0$, we get

$$0 = \ddot{X} = \cos \omega t [\ddot{x} - \omega^2 x \dot{t}^2 - 2\omega \dot{y} \dot{t}] - \sin \omega t [\ddot{y} - \omega^2 y \dot{t}^2 + 2\omega \dot{x} \dot{t}].$$

Since equations (2.85) hold for all t , this implies

$$\ddot{x} - \omega^2 x \dot{t}^2 - 2\omega \dot{y} \dot{t} = 0 \quad \text{and} \quad \ddot{y} - \omega^2 y \dot{t}^2 + 2\omega \dot{x} \dot{t} = 0 \quad \checkmark \quad \blacksquare$$